

ActInf MathStream 005.1 ~ Cristian Bodnar

"Topological Deep Learning: Graphs, Complexes, Sheaves"

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hello and welcome it's July 17 2023. we're here in active inference math stream number 5.1 with Chris Bodner on topological deep learning graphs complexes and sheaves so thank you for joining Chris looking forward to your presentation and discussion yeah thanks thanks all for uh having me so yeah as I was just saying um this is um my my PhD pieces which I've finished a couple of months ago it's also kind of a public online so you know if you want to go into the details just kind of look this up on the internet and uh you should be able to find it easily uh obviously there's lots of stuff in there so I kind of try to give an overview uh today and maybe also go in a little bit more detail in certain aspects um since there's you know not a lot of time to go through everything um so yeah now I'm on my Microsoft research so this is some basically all work that I did in the past then when I was at the University of Cambridge um all right so let's let's get uh started um right so let's let's start very easily um now I'm actually not sure exactly what's kind of the background of the people who are watching uh but uh in machine learning there's uh there's all these kind of subfield that emerged a few years ago which is called geometric deep learning which is essentially looking at how to apply this kind of deep learning neural network architectures on data living on all sorts of you know kind of structures or geometries or or spaces if you want um and this has a lot of applications especially in the Life Sciences and there's kind of been a lot of instances of this in kind of you know very famous Publications and all which you see here but you know just to give some examples for instance if you if you have proteins or molecules or things like that they usually represent it as graphs um and you kind of have some data living on these graphs like kind of the properties of certain atoms and and so on um so so these kind of things um and so far this kind of species or this kind of problems learning problems if you want they have been approached mostly kind of with a geometrical mindset as the kind of name of the of the subfield um also mentions um but um something that you know I would argue is that geometry

is not everything that you need and there's kind of other non-geometrical aspects when you are in such a setting and this is kind of quite obvious once you realize that the spaces that kind of show up in in the field and in many applications uh they are very heterogeneous so as I mentioned for instance you could have graphs that you know could represent anything in this case it's the caffeine molecule that you see here on the left and you want to you know have some models that predict certain properties of this molecule and so on uh but for instance you can have grids and we see grids all the time and data living on grids and I'm referring to images videos they are all kind of you know pixels living on a grid um and then you can have more sophisticated things you could have some meshes for instance uh they're all over in computer graphics and then you could have some sort of manifold so for instance if you're doing maybe weather modeling or something you you know we live on a sphere topologically speaking um so you might want to model your data as living on a sphere and and so on but nonetheless uh even if these spaces are kind of geometrically kind of heterogeneous and some of them not don't even have a geometrical structure in kind of a strict mathematical sense they all have What's called the topological structure which is kind of like kind of a weaker kind of structure uh but it's it's kind of more General um and I'm going to talk a bit uh in a few seconds about what that means but in general you do kind of mathematical physics you kind of have a ladder of structures where kind of uh you know you kind of keep building on top and the more structure you have the more sophisticated things you can do and so on and you know at the basis of this diagram just have sets it's just kind of a collection of elements with no kind of extra structure uh and then you kind of keep going up in this uh ladder and you add stop stop on top of sets and so um and as I was saying kind of most of the work is kind of focused on maybe the top levels of this hierarchy but it's kind of topological level it's a kind of uh you know the weakest kind of level you can add on top of that has kind of been neglected to a large extent um and part of what I've been doing in my PhD thesis was essentially looking at this kind of learning problems on these kind of spaces from a more topological perspective and you know kind of Fight The Filling uh these blanks um so this is kind of the overview and now okay so if we are to adopt this topological perspective what would that actually mean or how would that look like um I guess it could look in different ways but in what I've done in my faces it looks kind of like this so we have horizontally we kind of have a space um that could be anything it's just kind of an abstract space you know could be a grid or things like we've seen uh in the previous example and kind of vertically attached to the regions of the space we have data um so data is kind of this vertical component and you kind of see these uh

Flags kind of being you know kind of anchored in these regions so that kind of signifies you have some data that's kind of associated with that region um so that's kind of the high level perspective um and I'm going to make that a bit more concrete a bit later um and there's kind of two essential things about this picture uh first thing is locality so the data is uh attached to some regions of this space this topological space um and and in that sense it's local so it's kind of associated with the region and maybe to give a concrete example if you kind of have a temperature sensor somewhere in space right you could think of whatever that sensor is measuring is kind of a property of kind of the immediate surrounding of of you know around that sensor so it's kind of describing some property of of a region in space um so so that'll be kind of a good example um and another kind of Axiom uh that that uh we're starting with is that um the space has structure um so the space has kind of it's kind of made up of various regions and these regions intersect in various ways so the and and that implicitly also makes our data structure because the data is attached to these regions so so there is kind of some some structure in in the data um all right so that's kind of the picture and actually many of these things relate to category Theory I'm not gonna go in depth into this because it's kind of sophisticated and I'm not an expert myself in category Theory um but kind of the the high level here is that category theory is kind of a nice way to uh translate between different structure in mathematics and kind of you know discuss about properties of certain kinds of objects and translate that to some different kinds of objects and find all these kind of relations and connections and a concrete example is for instance if you want to study some manifolds on surfaces you could associate some groups to these surfaces and then any sorts of relations between this kind of surfaces also translating some relations about these groups or some other algebraic structures so you could study these manifolds by doing algebra instead of doing you know geometry or topology or something else um so also in this case this kind of manifests in the fact that we kind of translate these um from spaces to data so kind of because we associate the regions in in a space certain kinds of data this is kind of how these translation manifests in in what I've just described in the previous picture so you could think of this as some sort of translation or mapping from from spaces and regions in that space to kind of data attached to that space um but yeah I'm not going to go in a lot of detail into this but it's just kind of to you know to keep in the back of your mind um that there's there's this stuff lurking in the background um all right so this is I I promise this is the early math definition I'm giving in this talk and then and then I'll stop um but just because I'm mentioned in topological space spaces quite often I just wanted to kind of um give this axiomatic definition um which might sound

sophisticated but I have a picture at the end and hopefully it'll be clear um so yeah so what is it um so it's just a set as I was saying we start with set and we put stuff on top right so you start with a set and then you also have a collection of subsets of this set um called the open sets that need to satisfy a certain axioms so you could think of these open settings kind of regions of the space very informally speaking um so something that uh kind of has to be satisfied is that well the empty set and the set itself um need to be open set so in in some sense you could think of this as saying the set itself is a region of that space uh very informally um which you know it's a kind of uh say obvious and then there's some some kind of constraints about intersecting and taking unions of this region so uh if we take the intersection of two regions we should get another one of these regions um and if we take a union of these regions we should take we should get another region um and uh there is some constraints again like okay how big this intersection should be there should be finite intersections but you could have infinite unions but that's a technicality and we can just skip anyways but to see a picture so on the left you just see the set X itself and here I put like a potential neighborhood structure kind of like open set structure on this space so we have an open set here another open set V by this Axiom the intersections would also be an open set so yes it is intersection in the middle being another open set and then the set itself is another open set so it's just kind of a splitting stuff into regions kind of um you could think of it like that um all right so so this is this is a topological space um and now let's add data right so we we mentioned that we have data and then we put we sorry we have a space and then we add data on top uh so so far we've seen how a topological space looks like now let's add this kind of vertical stuff these flags that you saw before we we just put some data on top of these um and if we put data on all the regions of the space on all the open sets um we get these uh structures that in categories here like in algebraic topology also geometry they are called pre-shapes um which which sounds very fancy but all it's just kind of a definition of uh what I was already describing is actually you have some data for each region these these F of so for instance for uh bridge and U you have this F of U which is kind of the data attached to region you so you can think of f of U some some set with with describing the data that lives there in that region um but there's also an extra thing you you have some some sort of maps going between these kind of uh pieces of data um and these are called restriction maps and and why is that it's kind of they provide you a way to uh kind of zoom in if you want right like you have you have the data attached to the whole set X and then you could think okay how do I take this data how do I go from this data to data on a smaller region on on X so it's kind of a way to zoom in on that data in uh essentially and I'm going to

show some example in a second so these are called sheaves and just to see an example our space here is kind of one of the simplest kind of space you could think of just 1D or horizontal line right this is the real line and then you have some regions which are just given by open intervals uh on on these uh uh yeah on the real line and then some pieces of data could be functions like continuous functions on on those regions so here is like some sample data on on this first interval here's another function on the second interval here's some data on this third interval and actually in this case it happens that all these um functions agree on the overlap so where these regions overlap they take the same values and we can actually glue them together on a single uh in a single function over the entire region right so so this is just an example of um of a sheaf and it's called the sheaf of continuous functions um so our data in this case is continuous functions and the space is just the real line um and we put these functions on top to realign um and but it turns out because of this kind of special property uh that we can kind of glue data and we uniquely get some other piece of data right we we can take these three pieces it's just exactly like a like a puzzle right we put this thing through things together and we get a fourth thing which is kind of a single function uh where we just overlapped these these functions right um and uh these uh sheaves that satisfy this kind of properties where you can kind of glue them to get a unique piece of data they are called sheaves um and uh basically the sheaf of continuous functions is actually a sheaf um so this is kind of a way to formalize data attached to these things um it's gonna get less technical in a second um so just to give more examples so for instance in this way we could describe data on a sphere and let's say if this data is just some Vector field on this here so let's see if this is earth right this could be some wind Vector field right like if we do weather modeling or something you just have a vector field describing the wind on on the surface of Earth right um and you might want to do some machine learning on top of this where this kind of vector field has a sheaf structure and you could think of it as a sheaf because you know if you have like some Vector field on the red region a vector field on the yellow region I can kind of glue them together uniquely if they overlap if they agree on the overlap and we get the vector field on this bigger region but something that's quite nice is that even if we have a very very different kind of space namely a graph which is very different from a sphere in in all points of view and we can still apply the exact same kind of axioms and terminology and kind of definitions and we can have a sheaf over a graph so in this way we could have for instance some features associated to the nodes of the graph some features associated with the edges of the graph and there's another node which has its own features and this is actually the exact same thing

we have in graph machine learning um so this is quite nice for this kind of topological perspective allows us to do is you know we we kind of have a unified way of thinking if you want about very kind of heterogeneous spaces and we can model on all of them data attached to them by using this kind of shift terminology and other ways as well but I'm not going to that in this talk um all right so this is kind of an overview of what I've been doing in my thesis um and just to kind of dive a bit deeper into this um I just yeah I wanted to go in uh into one paper that we did at Europe's uh last year yeah so this was last year um on uh what's called shift diffusion so essentially how can we use what I've just described to do some useful stuff when doing machine learning on graphs um all right and this was a collaboration with Francisco de Giovanni but in Chamberlain uh Pietro Leo my advisor and Michael Bronstein um okay so before I dive into this I just want to give some background in case people are not familiar with this um so the kind of favorite architecture of people doing machine learning on graphs these days are these things called graph neural networks um which are actually very simple kind of models so in this setting you have some features so each node in your graph will have some features this is what is h Vector here denote so it's um the vector H associated with node a at layer or time D whatever so you have some features for each of these nodes and what you're doing each of if you are at a certain node you want to kind of compute a new representation or a new new features for for this now that the next layer so since you're learning representations um and what the graph neural networks are doing this node will receive a message from all the other nodes that are neighbors with this node um and this these uh message can also be passed you know can use on neural networks in there but essentially some processing of these features of the neighbors and these are aggregated into this message so here in green um and then there is this is passed through some update function um that combines the message from the neighbors with the old representation of this node and it gives you a new representation kind of the next layer um and this happens for all the nodes right so then you get some new representations for this node and this is one layer and then you kind of keep repeating this for as many layers as you um so this is kind of how you do deep learning on graphs um it's kind of a very very simple uh recipe and most models actually vary in the way they kind of compute these messages messages and in the way these update function is designed uh but that's kind of the parameter is the most of these models use otherwise they all kind of respect this framework and work in this kind of particular way um and to give you maybe an example for why you would want to do this uh you might want to do node classification this is kind of a classic problem in graph machine learning there are others but

I'm just going to talk about this because it's easier um so you have a graph and this graph um has nodes that have different labels here there's just two kinds of labels these orange and blue and you have some edges between uh these nodes and what you want is you want to do this kind of a message passing that I was describing to compute some representations for these nodes where you can easily classify you can easily classify the blue and the orange nodes um now something that's quite interesting is that for many kind of graph neural networks depending on the properties of these graphs and how these kind of different nodes are connected their performance might vary quite a lot um so in particular they're affected by this property called heterophily um so this kind of a measure of how much uh opposites attractive you want right so uh it has a very simple formula um basically you take the number of edges between orange and blue nodes and you divide by the total number of edges right so so basically you kind of check how many connections we have in this graph between things that are quite you know they are opposite to each other versus you know connections that are between similar kind of nodes um so if you have this kind of a lot of these kind of heterogeneous connections then you have very high heterophily um and it turns out that many Graphs are not exactly struggle in that setting it's very hard to to classify things in that setting um and iteratively you could kind of also you know figure out why because you know you could easily apply to some kind of reasoning where you know oh this node looks a lot like these other node is connected to so they kind of must be in you know in the same Community if you want or in the same label but it's much harder to do that when you know all things are kind of different from um from each other um and and this community is kind of don't form right even visually if you if you see a graph and it has a nicely clustered communities you know it's quite easy to draw a line between those and say oh this is uh communities another Community but if things are very mixed um then it's quite challenging and it turns out it's also challenging for these models uh not just for kind of our intuition when when you know we would have to do this um so this is kind of some problem where um this perspective this topological perspective I was mentioning uh will be used to to you know do some useful stuff okay so coming back to graphs on graph um and at this point I think you can largely forget what I mentioned in your introduction uh or if there's something give misunderstood there uh you know we kind of start from zero bit here so um there's no problem so on the left um you just have a graph um which is kind of the incident structure of a graph I just don't hear the simplest possible graph that has two nodes V and U and then there's an edge between them so this is just a graph with one Edge then that's all that's going on here and I've

just represented it by kind of in this kind of incidence structure kind of way right node V is incident to node e and node u sorry Edge e and node U is incident to h e so this is just an incidence structure and what is kind of triangle symbol we're showing is just these incidence structure is just the way to symbolize this incidence relation if you want um okay so this is just a graph right and a way we can kind of think of of fields on graphs is just mapping these graph structures so this is kind of this categorical Theory translation we translate this graph into something else which looks you know very similar the structure is kind of the same it's just kind of the meaning of these uh things change so for each node V we have um here this will be a vector space so F of e is a vector space for each node here we have F of U which is another Vector space uh for each Edge e we have this F_e which is another Vector space um so all nodes have their own Vector spaces and the features Associated to those nodes leaving those Vector spaces so basically for each node we have a vector space of features that that's all that's going on so far and also these arrows that these incident relations also translate into something and they translate into the obvious thing linear Maps so if these are vector spaces then these things should be linear Maps or just some matrices essentially right so for each Arrow you see here we have a matrix and uh see yeah and something uh that I'll I'll argue and show in a few slides is that basically message passing uh is very similar on graphs it's very similar with group actions in in group Theory um so let me explain exactly why so we kind of can think of the what we have on the left the these arrows in from the incidence relation they could think of these errors as kind of uh some patterns we can press so what do I mean by that so if we have this node V on the left right and this e now if we have some features some feature living in F_V right we could just kind of press this arrow button here and then if we multiply this matrix by this feature we will get an edge feature um so it's kind of like um if you go along this Arrow this Matrix will multiply this feature and this vertex feature and it will give you an edge feature um so if you think of these arrows as kind of giving you some sort of actions that you can play with to move features from vertex to Edge and and Edge to vertex so in this case it's kind of a left action right so this is what I'm saying I'm taking this R which is uh this one here and I act on some features of node V so this H_V that lives here right um and how I do that is I just take this Matrix this Matrix associated with this arrow and I multiply this Vector H_V so just Matrix times Vector that's all and then we get an edge feature so this is just kind of a way to to move from here to here so this already kind of looks a bit like message passing right we're kind of passing a message from this vertex to this Edge um but now we also need to pass a message from this Edge to these other vertex used so we kind of you know we need to get

from V to U um so and we did that by passing through e so by doing that we could do that by kind of going in reverse so we could have a right action um where instead of applying this Matrix we applied adjoint Matrix so it doesn't the transpose Matrix so in instead if we want to go from here to here um instead of applying this Matrix we apply its transpose because we want to go the other way around um so this is this is a if we compose these things then we can move features from V to U so this is just the way uh that kind of we can apply these actions uh to to do message passing and these are called shift actions or pre-shift actions and uh I'm gonna now say what's kind of the relation with between this and what we have in group uh Theory um so one way to represent the group is by kind of having some sort of graph um like here on the left so uh we kind of have some star object it is just kind of a dummy thing there uh but all the group structure is in these arrows right so for each group element so let's say this G is a 90 degree rotation for instance that that's uh let's say we do have a group of rotations that just to have some concrete example these uh so these Arrow could correspond to a 90 degree rotation we have another arrow that does the opposite minus kind of 90 degree rotation that's the inverse of that transformation um so this is kind of the structure of this group and if we have we also do this kind of similar kind of translation as we've just seen um so basic basically we Define a pre-shift on this group uh we map this star to a vector space so the star kind of replace is the vertex uh we had before now we just have a single vertex uh and it's just these arrows we have um so the star is not this Vector space that you showed here in blue right um and now if we actually do group actions which are kind of a very well established Concept in group Theory well for instance if you want to act on this vertex oh sorry not vertex on this vector v right here you have a vector in this Vector space and you want to act on it by this group transformation G so essentially you want to press this Arrow so you you apply some action on it um then what you do is well because of this translation this G has been mapped to some Matrix which is the rotation Matrix the corresponding rotation Matrix and you apply this rotation Matrix on V and you get like a 90 degree rotation here um so this is what what's going on this kind of vertical Vector is showing the rotated Vector so this is completely analogous with what we've seen on the previous slide um is uh this is how kind of chiefs connect these kind of actions so essentially what you could think of as message passing is same as group actions in group Theory but you just replace this group with a graph so it's kind of analogous to that right so uh it's just kind of a different kind of translations where we replace the object on the left uh now it looks like this is a graph this is a group right so but kind of the rest stays exactly the same so um you could and this kind of gives us a way to

formalize in a way by looking in this topological perspective to connect all these kind of symmetries and things like that that have been explored quite a lot in machine learning through message passing on graphs and to see one way in which in which they are related um okay so now in my circuit this was all very sophisticatedly and nice but you know what what is this going anywhere basically um and I'm just going to show you kind of a very forward example there there's more about you know the time is limited um and something we we showed is that uh as I was saying in the beginning many graphene networks kind of struggle in these heterophilic graphs and what we've shown is that um no matter how heterophilic or kind of weird your your graph is uh you can always kind of find some shift structure essentially kind of a message passing neural network that if you use sufficient layers it will be able to disentangle the classes of the of the nodes right so just to show you in this picture what you have here on the very far left um the colors of the nodes they show the class so we have three there's three colors here so three classes right um and this is kind of the graph in the beginning and the position of the nodes in this box denotes the features so that's a way to kind of just to visualize the features um the the position the 2D position is actually the 2D feature Vector of each node um and you can see in the beginning everything is kind of super messy and entangled right like if you want to classify these nodes it's kind of very hard because the representations the initial representations are very messy and kind of uh intertwined um but as we stack more layers of a particular kind of key for a message passing model um you see how progressively these classes get you know kind of more decent angle and more at each new layer and so that kind of this representations kind of collapse and they form this kind of clusterings right and then when you get with something like at the end you can kind of see these three communities very clearly and it's extremely easy to separate um and kind of the essence behind this results was we we showed for different kinds of problems what sorts of Chief or message passing models you need by using the theory to solve kind of problems and um this is quite important because it kind of shows you some important bits and pieces in the architecture that you might want to to kind of change or or use in order to solve certain kinds of problems and we also had some sort of impossibility results so if you use a graphing a network of some kind you can't solve this problem or you'll struggle to solve this problem and we also saw okay if you use some more general ones then you might have a chance so this is kind of some the you know high level view behind um these theoretical stuff um and what we're actually doing in practice um is to essentially learn these message passing functions or to learn the sheet or these matrices so in in practice like when someone gives you a node

classification tasks is very hard to know beforehand what what exactly is the um the right C for the right message passing model uh to sort that task um and what we do is actually we learn that from data so we learned these matrices that do the message passing we learn them from data um by passing some using some neural networks which are shown here in red um and then you learn how to kind of transfer features between these Vector spaces and kind of move them around so this is just showing how these vectors which are features of these nodes and edges how they kind of moved around by depends going through the Via this matrices just some Matrix multiplications and okay so so that's kind of a high level here behind this model and we evaluated this on some kind of real world uh heterophilic data sets um uh where you you have to classify nodes based on kind of various communities or different kinds of labels and these data sets going from um right to left they are getting more heterophilic so in some sense more challenging for classic architectures and our models which are kind of inspired by all this stuff the dimension they score quite highly in in these benchmarks um and at the same time they also revealed some or Justified some various choices that other models in this space have done but maybe they were not so well Justified or maybe they had different kind of motivations we also managed to kind of show why various things they were already doing why they made sense from from the point of view of this kind of uh Theory um all right well that's it's all um all I had um yeah thanks for for listening and yeah happy to chat more about this and also have lots of backup slides in case you know uh depending on how far away you can Venture offer with these questions oh well awesome work thank you for the presentation for people who are in the live chat they can write some questions but there's many things I think we could talk about so I want to start with reading a quote from an abstract of the paper by uh vanderlar kudal and DeVries just to kind of ground this in the active inference context and and really justify why the message passing approaches that you are describing are helping in the active inference modeling so the paper it's two papers it's called realizing synthetic active inference agents and they wrote with a full message passing account of synthetic active inference agents it becomes possible to derive and reuse message updates across models and move closer to Industrial applications of synthetic active inference framework so how does knowing the message passing structure help reuse a model across different settings or like facilitate the legibility of the model right um so first of all I'm not super familiar with the kind of active inference literature um so you know you'll have to help me there a bit in in for for to Anchor maybe the discussions a bit more into that um but I think if I understand correctly the the kind of question uh you're you're getting at is basically

how how can kind of message passing help us generalize you know in kind of various kinds of settings or maybe you know from one graph to another and and things like that and um these these isn't kind of active area of of research how exactly this generalization is happening um but something you could notice where is something that um for instance was shown like these models are quite good at for instance plotting patterns or structures uh depending on how exactly you implement them but for instance uh you know let's say you have a triangle in your graph or you know that'll be kind of a simple structure right you you have a triangle or some other kind of kind of Gadgets in your in your graph like particular subgraphs right that might show up in in different kinds of uh you know various graphs the graphs themselves might look completely different from each other but these kind of patterns might kind of be re-emerging in multiple like local patterns like three emerging multiple graphs and that could help your way to kind of generalize right like you could see for instance if you have clicks they're super important and kind of when you do social network modeling and things like that because they kind of show this kind of um you know close group of friends right they all talk to each other so they can form a click like everyone's connected to each other right and then you you might be able to use that in general if another completely different social context um where you know the these agents are again kind of communicating in a similar manner or connected in a similar manner even if the kind of overall pattern is quite different um and it goes Way Beyond just kind of structural similarities because there's also features in there so there's combinations of kind of structural patterns and features that you know give you even more complicated patterns right like you might have a triangle but then also two of the features in this triangle are you know look in a certain way and one that looks in another way so so that gives you even more kind of refinement then and uh you know even kind of richer pattern detection abilities um so so you have essentially this you know ability to kind of spot spot patterns at multiple scales as well so you could see this happening at in multiple skills you could have patterns of patterns right you could have uh Community entire communities connected in various patterns and so on um and again it's kind of also research question how do you capture these hierarchical patterns and so on um you know in general you have to do more message passing if you want to capture things that are further away from each other because otherwise they can't talk to each other right um so yeah I don't know if that that actually answered your question or if I was kind of going in the right direction there oh it's great it brings up a lot of different cool ideas like like this patterns all the way down but totally agree I I think we can now perhaps explore some more specific connections to active inference

because hopefully the the um listenership or viewership of this it's kind of like a two-way street like some people may be coming from more of your background and then learning about active inference and generative models as a specific system of Interest the first time but also certainly for a lot of people in the active inference space like these methods coming from category Theory have only recently come up to I guess more prominence in Bayesian modeling at least where we are so it's kind of it's a cool connection making yeah um I think one of the biggest touch points off the bat was like you mentioned multiplying a matrix by a vector and interpreting that as an edge so just in the in the inference part of the generative model of about sensory observations we always talk about the thermometer observation and then the underlying hidden State temperature so that like exactly describes that case and that's why we can represent the active inference generative models the perceptual parts and the action Parts in terms of matrix multiplication it's why the Matlab code for generative models does look mostly like matrix multiplication and it can all be done explicitly that way so how does that feed like are there models that don't have this feature or what do we gain by having like all of our edges defined as um appreciative action with a matrix and a vector in this setting of agent generative models with perception and action like once yeah yeah any thoughts on that yeah I think essentially kind of the the graph structure is kind of telling you these things interacting in some way right so um there there's some communication between these vertices if you you know if we're kind of in a graph setting um and then kind of what the Sheikh is giving you or or any message passive model essentially is is expressing a way that in which way that connection should manifest in the model or in what way that connection should be used to process information um so in this case I was mentioning okay we you have like linear Maps because you could go on the type if your type of data or vector spaces then this transformation will be some sort of linear Maps but it doesn't necessarily have to be so for instance it could go to any nonlinear transformation right like if and this is what's happening in general in practice um you know if you if you have a neighbor the message coming from that neighbor could be modulated by any sort of transformation you you want so it could be linear it could be non-linear it could be something I don't know you you can specify it basically uh but essentially you could think of this as you have a structure level telling you who should communicate to whom um and then you kind of have some semantics that is kind of shift is adding on top uh saying how should these things communicate right like the first thing is uh who should communicate or what should communicate and and then the semantics we had on top essentially describe uh how should that communication uh manifest uh essentially very

cool I think that Maps like exactly to how we talk about the sparsity of variables in degenerative model so here the topology of the nodes in the graph that we want to do message passing on are going to be describing like the agent and the environment or like the generative model that includes perception cognition and action it's a lot of people have proposed different sparsity architectures for integrated modeling of perception cognition action so one example would just be like kind of Around the Clock like action influences environment environment influences perception back to cognition you could add a self Loop blanket and different kinds of connectivities and that defines the sparsity topologically which is where you showed the stack and and you were on the second and the third levels I think of the stack and then like What flows it has to be described how it actually what that edge does so what what is that what is that that is also being provided yeah yeah exactly and it could even go to the extreme where you know does that edge actually do anything so for instance if you uh have a matrix that's just the zero Matrix for instance Associated to that edge you're just going to multiply by zero or an item that gives you zero and it's kind of essentially pruning that edge right like I can't get rid of it I don't want that communication to happen but there's kind of there's also kind of this possibility where these kind of semantics they override the structural level where you say okay I don't need to communicate you know with this other agent person or whatever it depends on what this vertices actually mean and in what context in what context you are and then there's also the case where you could do some sort of selective pruning where um these Matrix depending on um so in in kind of linear algebra The Matrix has a kernel so it's uh all the stuff that that Matrix sense to zero so what vectors are sent to zero right but not everything will be sent to zero unless you're the zero Matrix so depending on the features of the neighbors you could also just send some of the neighbors to zero right and that kind of removes those Neighbors From the equation they just kind of get you know you you yeah they're not they're not factored in anymore so you kind of have this you know it's a way to get the sparsity I guess that that they were also talking about uh where you know certain only maybe a small subset of the inputs or kind of uh yeah only a subset of the features are actually kind of doing some meaningful stuff among the neighbors and everything else will be kind of zeroed up yeah that makes me think of the lasso regression which tries to set most variables of having an impact of zero so that a few hopefully important variables really pop out in the analysis but also there's there's newer techniques I guess of attention modeling and maybe waiting that isn't just like okay set five of them to one and then the rest and then the rest of them to zero like more nuanced um so I think that sparsity with the expressivity is basically the The Best of

Both Worlds because you do want to have a situation where there is an edge but the attention being paid to it is zero so functionally that doesn't have an update on the belief State even though the in principle The Edge exists and that's why we can model situations where like the agent believes they have impact in the world but actually just because the edge in principle exists doesn't mean that it has any given impact and so that allows like the articulation of these models where they factorize and keeps interpretable motifs in terms of just Little clusters of uh motifs here in our case describing the action perception and cognition types of systems of interest but people I believe already implicitly do this um like they will often add an adjective and refer to X kind of active inference so like deep active inference with a temporal Horizon sophisticated active inference with this kind of nesting and those are pointing to a given feature but of course those features as we're hoping should be composable and so this seems to be bringing tools that are even more General than just action perception modeling because they're at a lower level of abstraction than like any specific system of interest but where this work and kind of Timeless thinking around cybernetic systems come together through the active inverse generative model as a base graph it gets very exciting yeah yeah and maybe it's also something we're emphasizing here is that even if this kind of semantic level can get rid of some edges right by doing this kind of pruning something it cannot get rid of is the computation so something that kind of that structural graph level forces you to do it kind of tells you what should you spend compute time on right because like if even if you're going to decide to prune an edge you still need to decide that which you know takes compute time so you still need to look at all your neighbors if you are not right and and decide what to prove in or maybe you're done print anything or whatever but you have to look at every Edge and one way to look at this is uh the kind of the graph structure defines your computational graph or kind of a computational yeah a series of computational steps you have to execute and then the kind of the shift structure or the message passing model actually specifies what those steps are and you know in one particular way they they look exactly um and so yeah that's one point and yeah you also mentioned attention and actually yeah I'm glad you did because this is actually quite related and in in certain ways more General than attention and actually maybe going back to this slide it might be a nice way to um to see this so here basically what happens in attention um instead of learning these matrices that we learned here uh in attention you learn attention coefficients here so just learn a scalar that's the attention coefficient how much attention should I pay to essentially well you know this overall Edge let's say which will be just a scalar what we do is kind of a bit more complicated because you just learn how do I

transform these neighbors so it's kind of a whole Matrix rather than a single scalar uh but there's also some subtle differences but in in a in a follow-up work we did we also combined this with with attention and and went a bit more General and um that that also worked uh quite well but the kind of underlying idea is is very similar you want to modulate the way it transform information based on the information itself right so we have this kind of one level of um recursivity if you want or kind of yeah that we're also alluding to in that you know it happens in active inference where okay you're you're so if I'm not V right my neighbor you it has some features and based on these features which are XU I'm gonna find out the Matrix that will be used to process X here right so it's kind of very recursive and it's what happens with attention right based on the features of node U I'm gonna compute an attention coefficient that I'm going to apply to this feature over here right I'm gonna decide based on this feature how much attention should I pay to it um and here we decide how should I process it more generally uh in a linear way um so you have this kind of loopiness structure embedded in there awesome all bring up a few more points because I think there's there's so many great pieces so Toby Sinclair Smythe who we recently discussed his dissertation in live stream 54. introduced a term or Lisa phrasing the compositional cognitive cartography and so thinking about the compositionality of cognitive systems and I think what you're describing here with this notion that the mappings are more General than the kind of attention mechanisms known famously today that those represent like a lower dimensional special case of one kind of architecture makes me think about how the Bayesian graph is kind of semantic in principle and can have all of these nice categorical formalisms around them but then and and you can even build the connector to empirical data with the pre-sheaf and the Sheaf which may be news to even many empirical researchers doing data analysis certainly was for me but the message passing provides a rigorous translation from whatever semantic model is proposed topologically to an implementation procedure that can be planned for and executed in linear time or at least with definable characteristics so message passing plays a really important part in going from like the abstract what is possible to the implementations of any of these actual models and it does it in a really General way where is it accurate to say that we hope that implementation with message passing compatible generative models will kind of roll out better because we won't have some of the engineering challenges that less reusable abstractions might carry it's it's hard to say I think there there's also certainly some limitations to this paradigm as well so just kind of doing this kind of message passing things you were mentioning one thing is that it kind of scales up uh quite easily like linearly with the size of the graph but that also come at

the cost so there is certain results showing this has limit in expressivity um so if you actually want to go beyond this for instance you have to instead of just looking at pairs of nodes you have to look at tuples and and this kind of high order uh groupings of nodes um in in order to kind of get higher exclusivity there's all sorts of techniques to do that um but yeah and there's always this kind of tension between being more expressive and being efficient that you know will always be there in any sort of algorithm or or method so so it's kind of hard to say I mean it's definitely we can definitely say this is kind of not the ultimate solution let's say if you want to do things message passing in itself you know but you know maybe doing some sort of computations on on on graphs you know could could be yeah maybe also something that that's kind of maybe missing a bit in in kind of the graph email setting is you know the context where you know your your graph is uh your assume your graph is known um and you know you need to have some graph structure at least the sensible way to construct it right but for for many kind of more I don't know like more obviously I phrase that um I guess for Less clearly the fine things like okay if I'm an agent doing perception in the real world or something right like if I'm trying to create a graph of the world you know like um what's what's an object what do I create what what do I create a note for right if I want to have like one note or object and there's some connections between objects and things like that just I know it's like some some wild um the custom item if you actually want to do that but let's say you want right then there's also like all these kind of blurry things like what's an object and what's not an object what's kind of somewhere in between maybe you know do you is that a node is that so it's kind of like I'm trying to say that the graph structure is kind of very discreet right like it's kind of the node is either there is not there and Edge is there it's not there but then the world is kind of really fuzzy right so if if you use graphs as a as a model for your world then there probably has to be some decision to be made somewhere about these kind of fuzzy Concepts uh they actually translate in a concrete graph entity like an object an edge or whatever or not uh based on you know some kind of inference procedure um and I don't know if we did that or not it's kind of humorous as kind of as intelligent agents uh but that's you know kind of some some interesting thing to think about maybe you could also well maybe one way to solve that there's also kind of stuff like soft edges and things like that and in some way you know if you have attention coefficients it's a bit like that you know like if an edge has a weight of 0.001 or something it's almost like not being there but you know it's still kind of there so it's uh you know it's a bit of a soft graph architecture uh but so I guess at the edge level you can implement this softness but I think it's a bit harder at kind of the note level right like uh how

do you kind of model a node that's kind of there and not there um yeah there's just some random thoughts that's very interesting about the fuzzy object identification and kind of similarities and differences between nodes and edges even though in some ways they have some similarities too um or interoperabilities to one other point of contact was like an underlying hidden space that we understand topologically that projects a vector space at in from different places so that could be a vector of thermometer readings and we want to have a smooth path within the homeostatic range defined up to a boundary Point not saying that that's the structure of the world but a structure of a very heuristic and simple model might be to aim for continuity and have a defined hid in the state space that has continuity underneath and is able to emit vectors so that kind of brings some of these classifier type discussions that you brought up and like the kind of fundamental impossibility of geometric classification because you you are going to end up with gray zones whereas even if it takes a bit wise description you can separate the network so that gives an actual completeness measure and that allows measures like I mean amount of computational resources or in a more like statistically principled way like the Bayesian information Criterion so how many nodes should we have we should be on some trade-off front in some modeling space I don't know what to tell you it's a map not the territory and that's because that's more justifiable and so uh even lifelike organisms might want to self evidence staying emitting from a living state and so that provides a really simple graphical architecture to cybernetic systems and then active inference explores a lot of different more specific motifs within that broader blanket persistence picture and the path of least action so that's what enables the physics in that space and why these methods which as far as I understand are often used in quantum mechanics or being able to come together with active inference this way yeah um some something that comes to mind when you mention this I think there's also like a Recent avenir research in this area where people and it's again kind of generated by the fact that you don't know the graph before beforehand many times and I think kind of the old school approach was well you you construct it based on some rules right like you you're gonna say all these I don't know some things are similar I'm going to put an edge between them and you define similarly in whatever where you like and so on um and there was this kind of recent Trend where what you try to do is kind of latent graph inference or some people call it manifold learning if you think of the graph as some sort of manifold but this kind of very informally and essentially what you would do is like you would map whatever you try to learn the raw observations um into some latent space and that's where you actually construct the graph you construct the graph in the latent space rather than kind of

in in the you know raw space so that might be kind of a way to deal with fuzziness as well right like because then I guess you might lose some some of these kind of very concrete one-to-one mappings because you might learn some node in the latent space that maybe corresponds to three or four Concepts kind of uh mixed together in you know so there's all these kind of nice uh experiments with neurons in the networks kind of visualized and you know they learn maybe a kind of a mixture of Concepts like like if you see what actually uh you know activates that neuron is actually you know maybe a few classes or different kinds of things it's not necessarily a single thing so it could be something very similar here where you have some very entangled um representations that are kind of distilled in in this latent graph and then in a way at least in concept stay even if like kind of in the latent space that's still like kind of a very you know clear combinatorial structure with respect to kind of your raw observations um that that structure kind of can still kind of encode the fuzziness of the world to some degree because you have this kind of mixture of Concepts that got distilled in the same node or things like that or maybe some Concepts could be represented by multiple nodes uh depending on you know in what you see this Concepts there might be all sorts of variations or concept or points of view and so on so um I think you know kind of latent graph inference could be uh you know quite an interesting way maybe to address some of these um issues where we were discussing although I think uh it kind of died off a bit in in the recent year at least as far as I've seen there were a few you know slightly fewer papers um on the topic hmm well certainly the agents proposed latent structure of the world the causal structure of the world is just map not territory and so it enables maybe some of those core screenings um could you go to the slide where there was a mapping between a smooth sphere and then a regular geometric shape uh yeah let's see in this one yeah just wanted to make one one point see if you had any um comments at the heart of some of the relationships that you're describing and where you pulled back to and kind in terms of generalization helps us understand this relationship between the the sphere and the geometry and one place that that and the implications for data processing and all of the computational science areas is if you are preserving or learning or analyzing geometry but not topology or the other way around you might get these different like data set aberrations like you might have the topology of the coffee cup but it looks like something totally different and so what we would really want to do would be understand the relationship between geometry and topology because if we could understand it in principle like you have it on the left side and then in practice with the data scheme on the right side or or insert your own left and right side there then we'd be able to do

data analysis in a way that respected slash preserved both the topology and the geometry so it's like two compatible perspectives that have their different like strengths and weaknesses and heuristics and so understanding that relationship between geometry and topology and the implicit um spaces that geometry requires and so on that has tremendous use and it just in closing reminds me of Buckminster Fuller's synergetics which uses a close packing architecture and a tetrahedron Centric model of coordinates to find more continuity between surface area and volume and between the smooth surfaces and the great circles on them and like the points of connectivity on shapes so I think it's an incredibly deep area it really has fundamental impact inactive inference helps us think about our models in this way kind of like the inflated balloon and with the fuzziness and the architecture and the finiteness it really brings a lot to active inference and so I appreciate you sharing the work with us today and continuing to work in this way thanks out any last thoughts um yeah what what you mentioned I think it's been all over my thesis this kind of tension between between topology and and geometry and maybe what I want to emphasize is that you know I'm not saying kind of the the previous perspective of looking maybe more geometrically of things was wrong in any way and on the contrary actually there's lots of interesting places where these things intersective and in kind of this Chief uh paper I briefly went there like if you actually read the paper there's uh there's a lot of beautiful intersections actually my my main collaborative Francesco he he's a differential geometer so um he actually had lots of kind of inputs from from that side and um indeed I think you know we should try to use all these kind of layers of structure uh you know in the best way possible for our methods awesome all right thank you until next time thanks for having me